

4.5 Quantum physics

This section provides knowledge and understanding of photons, the photoelectric effect, de Broglie waves and wave–particle duality.

In the photoelectric effect experiment, electromagnetic waves are used to eject surface electrons from metals. The electrons are ejected instantaneously and their energy is independent of the intensity of the radiation. The wave model is unable to explain the interaction

of these waves with matter. This single experiment led to the development of the photon model and was the cornerstone of quantum physics. Learners have the opportunity to carry out internet research into how the ideas of quantum physics developed (HSW1, 2, 7) and how scientific community validates the integrity of new knowledge before its acceptance (HSW11).

4.5.1 Photons

Learning outcomes		Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>		
(a)	the particulate nature (photon model) of electromagnetic radiation	
(b)	photon as a quantum of energy of electromagnetic radiation	
(c)	energy of a photon; $E = hf$ and $E = \frac{hc}{\lambda}$	
(d)	the electronvolt (eV) as a unit of energy	
(e)	(i) using LEDs and the equation $eV = \frac{hc}{\lambda}$ to estimate the value of Planck constant h	No knowledge of semiconductor theory is required. HSW11
	(ii) Determine the Planck constant using different coloured LEDs.	PAG6